

Recommended solution

1. Separate horizontal and vertical quality

Store every observation as:

$$z_k = [lat, lon, h, \sigma_h, \sigma_v, v, heading, t, satellites, source]$$

where:

- σ_h : horizontal accuracy in metres
- σ_v : vertical accuracy in metres
- h : GNSS altitude
- t : timestamp

A poor σ_v should reduce confidence in elevation only. It should **not** automatically remove latitude and longitude.

Suggested initial policy—tune from your field data:

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Suggested initial policy—tune from your field data:

Condition	Recommended handling
Missing coordinates, non-monotonic time, duplicate/stale location	Reject this single measurement; retain it in raw audit storage
Large horizontal uncertainty	Do not append directly to track; hold/down-weight it
Large vertical uncertainty but acceptable horizontal accuracy	Retain the horizontal position; mark elevation as low confidence
EKF innovation is unusually large	Down-weight via robust EKF or temporarily hold the measurement
Elevation conflicts with DEM/barometer but horizontal fit is good	Flag altitude anomaly; preserve horizontal track candidate
Several invalid fixes in a row	Keep a buffered segment; bridge short gaps, split only after a timeout

ArcGIS exposes horizontal accuracy, vertical accuracy, velocity, course, timestamp, and additional source properties; on supported platforms those properties can include GNSS as the source and satellite count.

2. Use an adaptive 3D EKF

Use a local **ENU** coordinate system—Easting, Northing, Up—rather than filtering raw latitude, longitude, and altitude directly.

Recommended EKF state:

$$\mathbf{x}_k = [E, N, U, v_E, v_N, v_U]^T$$

Measurement:

$$\mathbf{z}_k = [E_{GPS}, N_{GPS}, U_{GPS}]^T$$

Build a measurement-noise covariance matrix independently for horizontal and vertical uncertainty:

$$\mathbf{R}_k = \begin{bmatrix} \sigma_h^2 & 0 & 0 \\ 0 & \sigma_h^2 & 0 \\ 0 & 0 & \sigma_v^2 \end{bmatrix}$$

This is the key correction: when vertical accuracy is poor, increase only the U -component uncertainty. The filter then trusts the latitude–longitude coordinates more than altitude, rather than discarding the full point.

If your mobile device supplies IMU data, use acceleration and gyroscope data in EKF prediction; otherwise, a constant-velocity model is still workable for a trekking or vehicle route.

3. Gate innovations, not altitude alone

Before accepting an EKF measurement, calculate the innovation Mahalanobis distance:

$$d_k^2 = (\mathbf{z}_k - \hat{\mathbf{z}}_k)^T \mathbf{S}_k^{-1} (\mathbf{z}_k - \hat{\mathbf{z}}_k)$$

where:

$$\mathbf{S}_k = \mathbf{H} \mathbf{P}_k^- \mathbf{H}^T + \mathbf{R}_k$$

Decision rule:

- If d_k^2 is small: perform a normal EKF update.
- If it is moderately large: use a **robust update**—inflate $\mathbf{R}_k$, apply Huber weighting, or reduce measurement gain.
- If it is extremely large and horizontal accuracy is poor: quarantine the raw sample, but do not immediately delete surrounding track points.
- If horizontal coordinates are good but only altitude is abnormal: update E , N , while down-weighting or skipping U .

This prevents the “one bad altitude creates a hole in the route” problem.

4. Make elevation terrain-aware

For each filtered position, fetch terrain elevation from an offline or online DEM/elevation service and compare it with your GNSS/barometric altitude.

Instead of using fixed altitude change rules such as “reject if elevation changes by more than 10 m,” calculate a slope-aware residual:

$$r_h = |h_{GNSS} - h_{DEM}|$$

Then use a dynamic threshold:

$$T_h = \max(T_{min}, \alpha\sigma_v + \beta \cdot d_{horizontal} \cdot |\tan(\theta_{terrain})|)$$

where:

- T_{min} is a baseline elevation tolerance.
- $d_{horizontal}$ is movement since the preceding point.
- $\theta_{terrain}$ is local terrain slope.
- σ_v is vertical accuracy reported by the device.

This permits a larger genuine height change during steep climbs, while still detecting implausible vertical jumps on flat terrain.

Use the following hierarchy for elevation:

1. Barometric altitude after calibration, if the device provides it.
2. DEM-derived ground elevation as a terrain-consistency reference.
3. GNSS altitude only when vertical accuracy is acceptable.
4. Smoothed elevation for GPX export.

If altitude is important to your project, do not overwrite raw altitude—store both raw and filtered values.

Avoid gaps in the GPX trail

Use a **continuity-preserving track builder** after the quality gate and EKF.

- Keep a short rolling buffer, for example 5–15 seconds of observations.
- Do not create a segment break for a single rejected observation.
- If the gap duration is short and the distance/speed are plausible, interpolate one or more internal points between the last accepted and next accepted filtered point.
- Create a true GPX segment break only when GNSS is unavailable for a meaningful duration, the implied speed is impossible, or the user intentionally pauses/stops tracking.
- Preserve “protected” vertices: start, end, sharp turns, quality-state changes, first/last point around a signal loss, and terrain-transition locations.

Your map can use the ArcGIS location display for the live device marker, but your GPX line should be rendered from the separately maintained filtered-track list. ArcGIS notes that the location display represents reported accuracy and that accuracy changes with signal and other conditions; therefore, the live location symbol should not itself be treated as a validated trail record. 

For example:

Terrain / route condition	Horizontal simplification tolerance	Vertical tolerance
Switchback, ridge, crossing, sharp turn	2–4 m	2–5 m
Normal walking trail	4–8 m	5–10 m
Smooth road-like section	8–15 m	10–20 m
Poor vertical GNSS accuracy	Keep horizontal geometry; do not use raw altitude for removal	Use filtered/DEM-supported elevation

GPX record design

For every exported `<trkpt>`, retain the quality context in GPX extensions rather than deleting a point merely because its altitude confidence was weak:

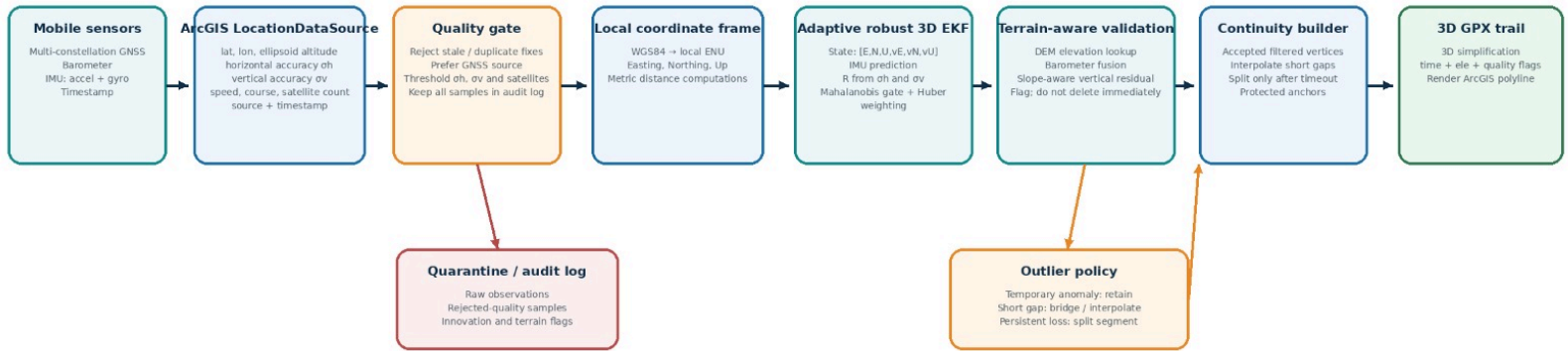
xml

```
<trkpt lat="18.986245" lon="73.124784">
  <ele>642.3</ele>
  <time>2026-09-03T08:52:14Z</time>
  <extensions>
    <horizontalAccuracy>4.8</horizontalAccuracy>
    <verticalAccuracy>16.5</verticalAccuracy>
    <quality>VERTICAL_LOW_CONFIDENCE</quality>
    <filter>ROBUST_3D_EKF</filter>
    <terrainCheck>FLAGGED_NOT_REMOVED</terrainCheck>
  </extensions>
</trkpt>
```

This gives you a clean visual trail while preserving traceability for research evaluation and post-processing.

Robust 3D GPS Trail Processing Architecture

Terrain-aware filtering prevents false altitude-triggered deletions and broken GPX trails



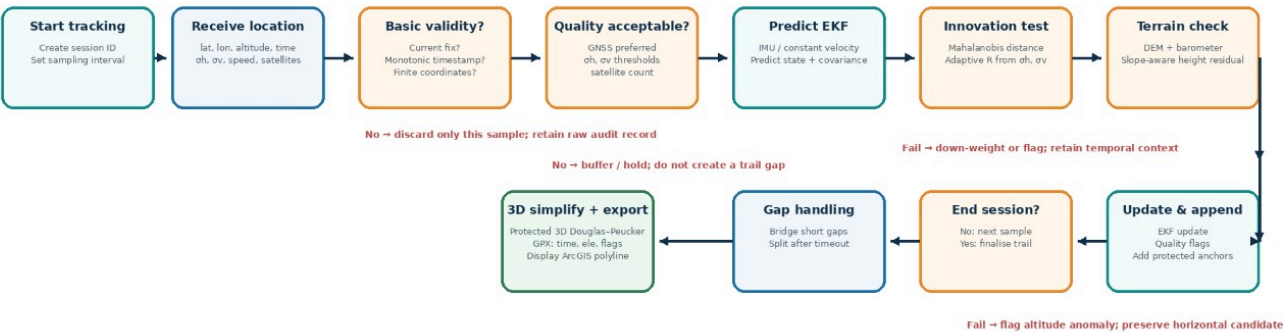
Recommended decision logic

1. Hard reject only invalid/stale fixes, extreme horizontal uncertainty, or impossible time order.
2. For EKF innovation failures, down-weight or hold the measurement; do not remove neighbouring valid points.
3. Compare altitude to local terrain using slope and vertical accuracy; steep terrain permits larger height changes.
4. Protect start/end, sharp turns, quality-state changes, and long-gap boundaries before 3D Douglas-Peucker.

Key principle: altitude informs confidence; it must not be a stand-alone reason to delete a horizontal GPS position.

Terrain-Aware GPS Track Processing Flow

Confidence, temporal continuity, and terrain context are evaluated before GPX simplification



Implementation safeguards

- Use anisotropic measurement uncertainty: horizontal and vertical GPS errors are not equivalent.
- A reliable latitude-longitude observation may arrive with unreliable altitude. Store it with low vertical confidence instead of deleting it.
- Run simplification only after filtering and gap handling. Preserve anchors at turns, start/end, state changes, and gap boundaries.
- Keep a raw audit log to tune thresholds from real mountain recordings and explain every rejected or down-weighted point.